Short-Term SAR Change Detection for Soil Moisture Retrieval: A Case Study Over Danish Test Sites

Miquel Negre Dou and John Peter Merryman Boncori

***Abstract*—This article investigates the use of a constrained short-term change detection (STCD) method for field-scale soil moisture (SM) retrieval from Sentinel-1 synthetic aperture radar (SAR) backscatter. This method estimates the changes in surface dielectric properties by analyzing backscatter ratios between consecutive acquisitions, under the assumption that SM fluctuates more rapidly than vegetation and/or roughness changes. The inversion scheme integrates ancillary data, namely, a coarse-scale SM product, soil texture maps, and field capacity, to constrain the solution and minimize outliers. The influence of key parameters, including the inversion window size, was examined. Additional corrections are introduced to handle limitations arising from crop type or measurement depth, including a seasonal adjustment model. The approach is validated using in situ SM data from multiple test sites in Denmark. Results show that the STCD method achieves strong agreement with ground-based observations, yielding an overall correlation of $R = 0.72$, low bias (−0.26%), and a root mean squared error (RMSE) of 4.32%. These findings support the potential of short-term SAR change detection for scalable, high-resolution SM monitoring in agricultural landscapes, where backscatter variability is well characterized.**

***Index Terms*—Sentinel-1, short-term, soil moisture (SM), synthetic aperture radar (SAR).**

I. Introduction

SOIL moisture plays a key role in a variety of applications, including agricultural monitoring, hydrological modeling, and drought forecasting [1]. Accurate and timely information on surface soil moisture (SM) can help optimize irrigation practices, improve yield predictions, and support water resource management [2]. A range of satellite-based products have been developed to estimate SM at regional to global scales, relying on both passive and active microwave sensors [3], [4], [5], [6]. However, these products typically offer spatial resolutions on the order of several kilometers, which limits their applicability for field-scale decision-making [7].

Radar-based sensors, such as those on board the Copernicus Sentinel-1 satellites, provide frequent and weather-independent observations, making them particularly promising for SM retrieval. Nevertheless, the standard spatial resolution of Sentinel-1-based operational products remains around 1 km [3], [8]. While suitable for large-scale assessments, this resolution does not capture the heterogeneity present across individual agricultural plots. Yet, several studies have highlighted the benefits of leveraging SM information in the backscatter signal, in combination with other sources, at higher spatial resolutions (e.g., 100 m or finer) for precision agriculture and localized hydrological applications [7], [9].

In this work, we investigate the feasibility of retrieving SM at field scale by validating the short-term change detection (STCD) method [3] against multiple agricultural sites in Denmark. We discuss some algorithmic choices and propose some refinements, which improve the level of agreement with in situ observations.

The remainder of this article is structured as follows. Section II details the methodology underlying the STCD-based retrieval approach. Section III describes the validation sites and associated in situ measurements used for performance assessment. Section IV outlines the input datasets and ancillary information required by the algorithm. Validation results are presented and discussed in Section V, followed by concluding remarks in Section VI. Finally, the Appendices provide a sensitivity analysis of key algorithm parameters.

II. Methodology

A methodological framework for retrieving surface SM from synthetic aperture radar (SAR) observations is presented, combining STCD with an electromagnetic forward model. The approach incorporates external constraints to resolve degeneracies in the inversion process and enhance retrieval reliability. The core aspects of the method are elaborated upon next, including backscatter modeling, polarization amplitude estimation, dielectric characterization, SM retrieval, and seasonal adjustment strategies.

A. Short-Term Change Detection

The normalized backscattering coefficient (σ_0) at a given polarization (PP) can be expressed as the sum of contributions from the soil surface ($\sigma^{\text{soil}}_{0,\text{PP}}$), the vegetation canopy ($\sigma^{\text{veg}}_{0,\text{PP}}$), and interactions between the two media ($\sigma^{\text{int}}_{0,\text{PP}}$)

$$\sigma_{0,\text{PP}} = \sigma^{\text{soil}}_{0,\text{PP}} \cdot \tau^2 + \sigma^{\text{veg}}_{0,\text{PP}} + \sigma^{\text{int}}_{0,\text{PP}}. \quad (1)$$

Received 4 July 2025; revised 12 December 2025; accepted 21 January 2026. Date of publication 26 January 2026; date of current version 11 February 2026. This work was supported in part by the Technical University of Denmark (DTU) Space, Department of Microwaves and Remote Sensing; and in part by the Danish Agricultural Agency (Landbrugsstyrelsen) through the KortDrænN2O project. *(Corresponding author: Miquel Negre Dou.)*

The authors are with the National Space Institute, Technical University of Denmark (DTU), 2800 Kongens Lyngby, Denmark (e-mail: miqne@space.dtu.dk).

Digital Object Identifier 10.1109/TGRS.2026.3658136

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

Assuming bare or vegetated surfaces where attenuated surface backscattering is the dominant mechanism, volume scattering and soil–vegetation interactions can be neglected. Under this simplification, σ_0 can be expressed in terms of the polarization amplitude ($|\alpha_{\mathrm{PP}}|$), the two-way vegetation attenuation (τ), and the soil roughness coefficient (Ω) as

$$\sigma_{0,\mathrm{PP}} \approx \sigma_{0,\mathrm{PP}}^{\mathrm{soil}} \cdot \tau^2 = |\alpha_{\mathrm{PP}}(\varepsilon_r, \theta)|^2 \cdot \Omega(f, \theta, \chi) \cdot \tau^2 \tag{2}$$

where ε_r represents the relative dielectric constant of the soil, θ is the incidence angle, and f is the carrier frequency of the radar, while χ characterizes the surface roughness spectrum.

This formulation, introduced in [3], is central to the STCD method, which relies on the assumption that changes in vegetation cover and surface roughness occur at a slower rate compared to SM fluctuations. Following this rationale and provided that the two radar images are sufficiently close in time, the ratio between their corresponding σ_0 values—denoted as $\sigma_{0,\mathrm{PP}}^{(k)}$ and $\sigma_{0,\mathrm{PP}}^{(k+1)}$, with k indexing acquisitions in the SAR time series—can be expressed solely in terms of the polarization amplitude

$$\left(\frac{\sigma_{0,\mathrm{PP}}^{(k)}}{\sigma_{0,\mathrm{PP}}^{(k+1)}}\right) \approx \left|\frac{\alpha_{\mathrm{PP}}^{(k)}\left(\varepsilon_r^{(k)}, \theta\right)}{\alpha_{\mathrm{PP}}^{(k+1)}\left(\varepsilon_r^{(k+1)}, \theta\right)}\right|^2 \tag{3}$$

which can be rearranged to

$$\left|\alpha_{\mathrm{PP}}^{(k)}\left(\varepsilon_r^{(k)}, \theta\right)\right| - \sqrt{\left(\frac{\sigma_{0,\mathrm{PP}}^{(k)}}{\sigma_{0,\mathrm{PP}}^{(k+1)}}\right)}\left|\alpha_{\mathrm{PP}}^{(k+1)}\left(\varepsilon_r^{(k+1)}, \theta\right)\right| \approx 0. \tag{4}$$

For this relation to hold, all scenes over a given location must be acquired with consistent viewing geometry—as is the case for acquisitions along a common ground track—or, alternatively, the $\sigma_{0,\mathrm{PP}}$ measurements are to be adjusted to a common reference incidence angle, following the approach in [10]. Here, the former course of action is adopted: separate per-track moisture inversions are performed and then merged onto a shared temporal axis, thereby avoiding cross-angle normalization. Additionally, the revisit time should be shorter than the characteristic timescale of vegetation and surface roughness change, increasing the likelihood that $\tau^{(k)} \approx \tau^{(k+1)}$ and $\Omega^{(k)} \approx \Omega^{(k+1)}$.

In the following steps, the analysis is restricted to vertical transmit, vertical receive (VV) polarization, as it has been shown to provide the highest sensitivity to SM among the available Sentinel-1 configurations compared to, for example, vertical transmit, horizontal receive (VH), which is acquired together with VV in dual-polarization products [11].

B. Alpha Approximation

Given a time series of N radar images, (4) can be rewritten as a system of equations, such that

$$\mathbf{A} \cdot |\vec{\boldsymbol{\alpha}}_{\boldsymbol{VV}}| \approx \mathbf{0} \tag{5}$$

where $\mathbf{A}$ is an $(N-1) \times N$ matrix with ones on the main diagonal and the negative square roots of the backscattering coefficient ratios on the first superdiagonal

$$\mathbf{A} = \begin{bmatrix} 1 & -\sqrt{\frac{\sigma_{0,\mathrm{VV}}^{(1)}}{\sigma_{0,\mathrm{VV}}^{(2)}}} & 0 & \cdots & 0 & 0 \\ 0 & 1 & -\sqrt{\frac{\sigma_{0,\mathrm{VV}}^{(2)}}{\sigma_{0,\mathrm{VV}}^{(3)}}} & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -\sqrt{\frac{\sigma_{0,\mathrm{VV}}^{(N-1)}}{\sigma_{0,\mathrm{VV}}^{(N)}}} \end{bmatrix} \tag{6}$$

where $|\vec{\boldsymbol{\alpha}}_{\boldsymbol{VV}}|$ is a vector of size $N \times 1$ containing the unknown polarization amplitudes corresponding to each radar acquisition

$$|\vec{\boldsymbol{\alpha}}_{\boldsymbol{VV}}| = \begin{bmatrix} \left|\alpha_{\mathrm{VV}}^{(1)}\left(\varepsilon_r^{(1)}, \theta\right)\right| \\ \left|\alpha_{\mathrm{VV}}^{(2)}\left(\varepsilon_r^{(2)}, \theta\right)\right| \\ \vdots \\ \left|\alpha_{\mathrm{VV}}^{(N-1)}\left(\varepsilon_r^{(N-1)}, \theta\right)\right| \\ \left|\alpha_{\mathrm{VV}}^{(N)}\left(\varepsilon_r^{(N)}, \theta\right)\right| \end{bmatrix} \tag{7}$$

and $\mathbf{0}$ is the null vector of the same size.

The system in (5) contains N unknowns and $N-1$ equations, rendering it underdetermined with one degree of freedom. To constrain the solution, external information on the polarization amplitude—linearly independent of the equations already accounted for—is required.

Once $|\vec{\boldsymbol{\alpha}}_{\boldsymbol{VV}}|$ is determined, each element ($\alpha_{\mathrm{VV}}^{(k)}$, for $k = \{1, \ldots, N\}$) can be related to a dielectric constant through

$$|\alpha_{\mathrm{VV}}^{(k)}\left(\varepsilon_r^{(k)}, \theta\right)| = \left|\frac{\left(\varepsilon_r^{(k)} - 1\right)\left(\sin^2(\theta) - \varepsilon_r^{(k)}\left(1 + \sin^2(\theta)\right)\right)}{\left(\varepsilon_r^{(k)}\cos(\theta) + \sqrt{\varepsilon_r^{(k)} - \sin^2(\theta)}\right)^2}\right| \tag{8}$$

from which SM can subsequently be obtained if a suitable electromagnetic model is used [12].

C. Electromagnetic Model

The sensitivity of $\sigma_{0,\mathrm{VV}}$ to surface SM fluctuations is driven by changes in the soil's dielectric properties. To relate these quantities, empirical models, such as the one proposed in [13] for mineral soils, can be used. This model describes the dielectric constant of soil–water mixtures as a function of frequency (f), volumetric moisture content (Θ), and the soil's sand (S) and clay (C) fractions. It utilizes a polynomial fit with parameters derived from laboratory measurements

$$\begin{aligned} \varepsilon_r = \varepsilon_r' + j\varepsilon_r'' = (a_0 + a_1 S + a_2 C) + (b_0 + b_1 S + b_2 C)\Theta \\ + (c_0 + c_1 S + c_2 C)\Theta^2 \end{aligned} \tag{9}$$

where the coefficients a_i, b_i, and c_i are complex, allowing for the calculation of both the real (ε') and imaginary (ε'') components of ε. Fig. 1 presents the expected $\sigma_{0,\mathrm{VV}}$ ratios with respect to surface SM changes.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

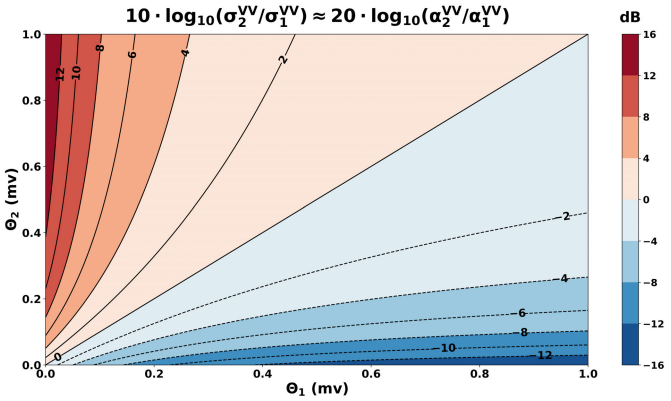


Fig. 1. Expected $\sigma_{0,\mathrm{VV}}$ ratio (in dB) between two consecutive SAR acquisitions as a function of surface SM changes (Θ_1 and Θ_2).

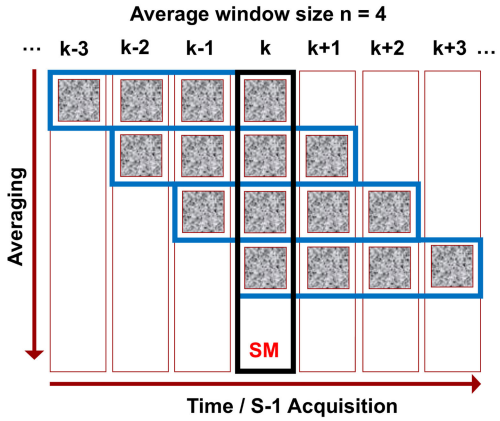


Fig. 2. Iterative retrieval algorithm using a moving window of size $n = 4$. Each gray image represents a Sentinel-1 acquisition, while the blue boxes indicate the inversion process applied to n consecutive acquisitions. The y-axis represents the averaging across all inversions associated with a single image, as highlighted by the black box for the central acquisition k.

D. SM Retrieval

The retrieval chain from backscatter observations to SM estimates can be outlined as follows:

$$\begin{bmatrix} \sigma_{0,\mathrm{VV}}^{(1)} \\ \vdots \\ \sigma_{0,\mathrm{VV}}^{(N)} \end{bmatrix} \xrightarrow{(6)} \mathbf{A} \xrightarrow[\text{information}]{\text{change + External}} |\vec{\alpha}_{VV}| \xrightarrow{(8)} \begin{bmatrix} \varepsilon_r^{(1)} \\ \vdots \\ \varepsilon_r^{(N)} \end{bmatrix} \xrightarrow{(9)} \begin{bmatrix} \Theta^{(1)} \\ \vdots \\ \Theta^{(N)} \end{bmatrix}. \tag{10}$$

While deriving SM from $|\vec{\alpha}_{VV}|$ is straightforward, obtaining accurate estimates of the polarization amplitudes remains challenging and requires external information, e.g., from coarse resolution SM products, such as those described in [8], [14], [15], [16], [17], and [18].

In principle, a single known element of $|\vec{\alpha}_{VV}|$ would suffice to recover the entire time series. However, to mitigate the influence of potential outliers in the external dataset, Balenzano et al. [3] proposed a sliding window retrieval strategy, illustrated in Fig. 2. In this approach, the subsets of $|\vec{\alpha}_{VV}|$ are estimated iteratively using a moving set of n consecutive images (from now on to be referred as time-window size). Although a bounded least squares approach was originally proposed by [19] to solve the underdetermined system (5), this is very sensitive to the accuracy of the boundary conditions [20]. We found more robust retrievals to be provided by imposing an additional constraint, based on the assumption of linearity between the normalized backscattering coefficient and the SM (see Appendix A). More specifically, for the jth time window, the SM value corresponding to the last date, Θ_0, is derived from the estimates available from the external dataset, $\vec{\Theta}_{\mathrm{ext}}^{\mathrm{win}}$, and the backscatter values, $\vec{\sigma}_{0,\mathrm{VV}}^{\mathrm{win}}$, based on the following equation:

$$\Theta_0 = \left(\max\left(\vec{\Theta}_{\mathrm{ext}}^{\mathrm{win}}\right) - \min\left(\vec{\Theta}_{\mathrm{ext}}^{\mathrm{win}}\right)\right) \cdot \left(\frac{\sigma_{0,\mathrm{VV}}^{(j+n-1)} - \min\left(\vec{\sigma}_{0,\mathrm{VV}}^{\mathrm{win}}\right)}{\max\left(\vec{\sigma}_{0,\mathrm{VV}}^{\mathrm{win}}\right) - \min\left(\vec{\sigma}_{0,\mathrm{VV}}^{\mathrm{win}}\right)}\right) + \min\left(\vec{\Theta}_{\mathrm{ext}}^{\mathrm{win}}\right). \tag{11}$$

This is then converted to the polarization amplitude of the last date in the window, α_0, using (8) and (9). By imposing the additional condition that $\left|\alpha_{\mathrm{VV}}^{(j+n-1)}(\varepsilon_r^{(j+n-1)}, \theta)\right| = \alpha_0$, the system of equations in (5) becomes solvable for each window, allowing the corresponding SM time series to be retrieved.

To avoid nonphysical values in the inversion, the estimated SM is truncated at a predefined threshold. In this study, a modeled estimate of the field capacity—defined as the maximum amount of water the soil can retain after excess water has drained—was used as upper limit.

After processing all possible consecutive windows of size n, the algorithm averages the inversion results associated with each acquisition, as shown in Fig. 2, to reduce the impact of inaccuracies introduced by the initialization of α_0.

The window size (n) acts as a regularization parameter, introducing a tradeoff: smaller values of n yield estimates that more closely follow the coarser external dataset, thereby reducing the influence of potential backscatter fluctuations, but potentially missing the higher level of detail provided by the backscatter measurements. In contrast, larger values of n rely more heavily on the backscatter signal, enhancing the spatial resolution, but also increasing the sensitivity to noise and systematic biases. A quantitative analysis of this tradeoff is provided in Appendix B.

E. Seasonal Adjustment for Crop and Depth Correction

Although backscatter ratios help mitigate the impact of changing vegetation and surface conditions, the SM signal may still be obscured by other processes, which contribute to the additive terms $\sigma_{0,\mathrm{PP}}^{(\mathrm{veg})}$ and $\sigma_{0,\mathrm{PP}}^{(\mathrm{int})}$ in (1). Root and tuber crops, for example, which concentrate a significant portion of their biomass just below or at the soil surface, can alter the radar response through changes in surface roughness and increased volume scattering near the soil interface [21]. These effects can mask the seasonal variability in SM, making accurate retrieval challenging.

To minimize this limitation, we apply an additional seasonal correction to fields, where such crop types are present. This involves fitting an asymmetric sinusoidal curve, $S(t)$, to the annual SM cycle, approximating seasonal dynamics with two

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

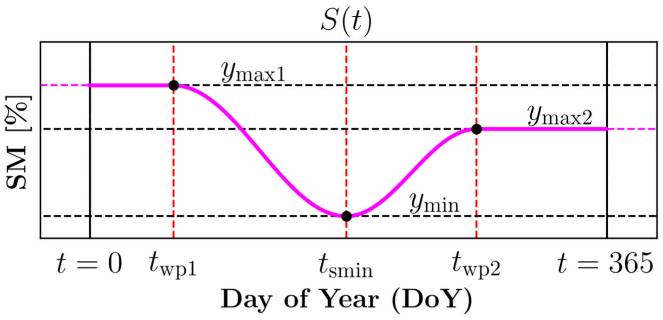


Fig. 3. Illustration of the seasonal adjustment curve, $S(t)$, for a given year.

winter plateaus, separated by a summer minimum, as shown in Fig. 3 and in the following equation:

$$S(t) = \begin{cases} y_{\max_1}, & 0 \le t < t_{\text{wp1}} \\ A_1 \cos\left(\pi \dfrac{t - t_{\text{wp1}}}{t_{\text{smin}} - t_{\text{wp1}}}\right) + D_1, & t_{\text{wp1}} \le t < t_{\text{smin}} \\ A_2 \cos\left(\pi \dfrac{t - t_{\text{smin}}}{t_{\text{wp2}} - t_{\text{smin}}} + \pi\right) + D_2, & t_{\text{smin}} \le t < t_{\text{wp2}} \\ y_{\max_2}, & t_{\text{wp2}} \le t \le 365 \end{cases} \tag{12}$$

where t is the day-of-year (with $t = 0$ being January 1st) and $t_{\text{wp1}}, t_{\text{smin}}, t_{\text{wp2}}$ denote the dates of the end of the first winter plateau, the summer minimum, and the start of the second winter plateau, respectively. The values $y_{\max_1}$ and $y_{\max_2}$ represent the SM levels during the two winter plateaus, while $y_{\min}$ corresponds to the summer minimum. The amplitudes and vertical offsets are given by $A_i = (y_{\max_i} - y_{\min})/2$ and $D_i = (y_{\max_i} + y_{\min})/2$ for $i = 1, 2$.

The first winter plateau spans from the start of the year until the latest day between October and March when the derived SM estimates remain within 5% of its seasonal maximum. Similarly, the second plateau start is selected in the subsequent October–March period as the earliest day satisfying the same criterion and continues until the end of the year, ensuring continuity with the next cycle. The summer minimum is set to the day with the lowest SM between April and September, based on the external, coarse resolution, dataset. This avoids using potentially unstable estimates during the crop maturity and harvest stages, when surface conditions can cause abrupt changes in backscatter.

The SM levels associated with the winter plateaus are computed as the 95th percentile of the uncorrected SAR-based predictions within the corresponding winter period, capped 5% below field capacity to avoid overestimates. The summer minimum is defined as the 15th percentile of the derived estimates over the April–September interval.

To align the SM estimates time series with this seasonal reference, a mean bias correction is applied using a 30-day centered moving window. Within each window, the local mean difference between the observed and reference series is estimated and subtracted from the observed values, effectively correcting short-term biases while preserving the seasonal structure.

Besides correcting for seasonal inconsistencies, this approach may also serve as a proxy for deeper SM dynamics. By further constraining the estimates with these seasonal patterns, it becomes possible to correct for temporal lags between surface moisture and that of deeper layers, while also reducing influences of shallow-layer processes that might not have an impact at greater depths.

III. Validation Sites and Field Measurements

Two distinct validation sites are considered in this study: an experimental field in Tokkerup and the SM monitoring network of the Danish hydrological observatory (HOBE). Sections III-A and III-B describe their main characteristics, including instrumentation, data collection protocols, and relevant field-specific conditions. These ground-based measurements will serve as reference for assessing the accuracy and robustness of the developed approach.

A. Tokkerup

The first study area is a 3.93-hectare (ha) agricultural field in Tokkerup, southeastern Zealand, Denmark (55°17′16.63″N, 12°8′24.59″E), shown in Fig. 4(a). This site was originally established to investigate the effects of different drainage scenarios on nitrogen dynamics, specifically focusing on nitrogen leaching and nitrous oxide (N_2O) emissions under varying SM and waterlogging conditions. These processes have been analyzed in detail by [22]. The field consists of sandy loam soil and was divided into two sections based on drainage performance: a well-drained (WD) area (2.65 ha), with tile drains installed at 90–120 cm depth, and a poorly drained (PD) area (1.28 ha), where the drains were placed at shallower depths of 55–70 cm, as described by [22].

In order to monitor soil water content, a CR1000 datalogger with a TDR200 multiplexing system was employed (Campbell Scientific, Logan, USA). Nine time-domain reflectometry (TDR) probes, 30 cm in length, recording hourly, were installed horizontally at depths of 30, 60, and 90 cm across three nests positioned at 50, 425, and 800 cm from the side drain. These recordings were carried out between June 2018 and December 2020 and are shown in Fig. 4(b). Due to the high bulk density of the soil at the study site, raw measurements were corrected using the calibration introduced in [23].

This setup allowed for continuous monitoring of soil water dynamics; significant differences in moisture retention between the WD and PD areas were primarily observed at deeper soil layers. While near surface SM levels (30 cm depth) showed minimal variation between the two sections, the PD area exhibited notably higher and more stable moisture levels at 60 and 90 cm depths [see Fig. 4(b)–(d)]. This was attributed to poor drainage and soil compaction in the PD section, resulting in prolonged waterlogging, increased denitrification, and greater nitrogen losses. In contrast, the WD section demonstrated better drainage at deeper layers, leading to lower SM levels and more consistent nitrogen uptake by crops [22].

The site is located in a temperate climate zone and was managed under conservation tillage, following an annual rotation of winter wheat (WW) and spring barley (SB), with no

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

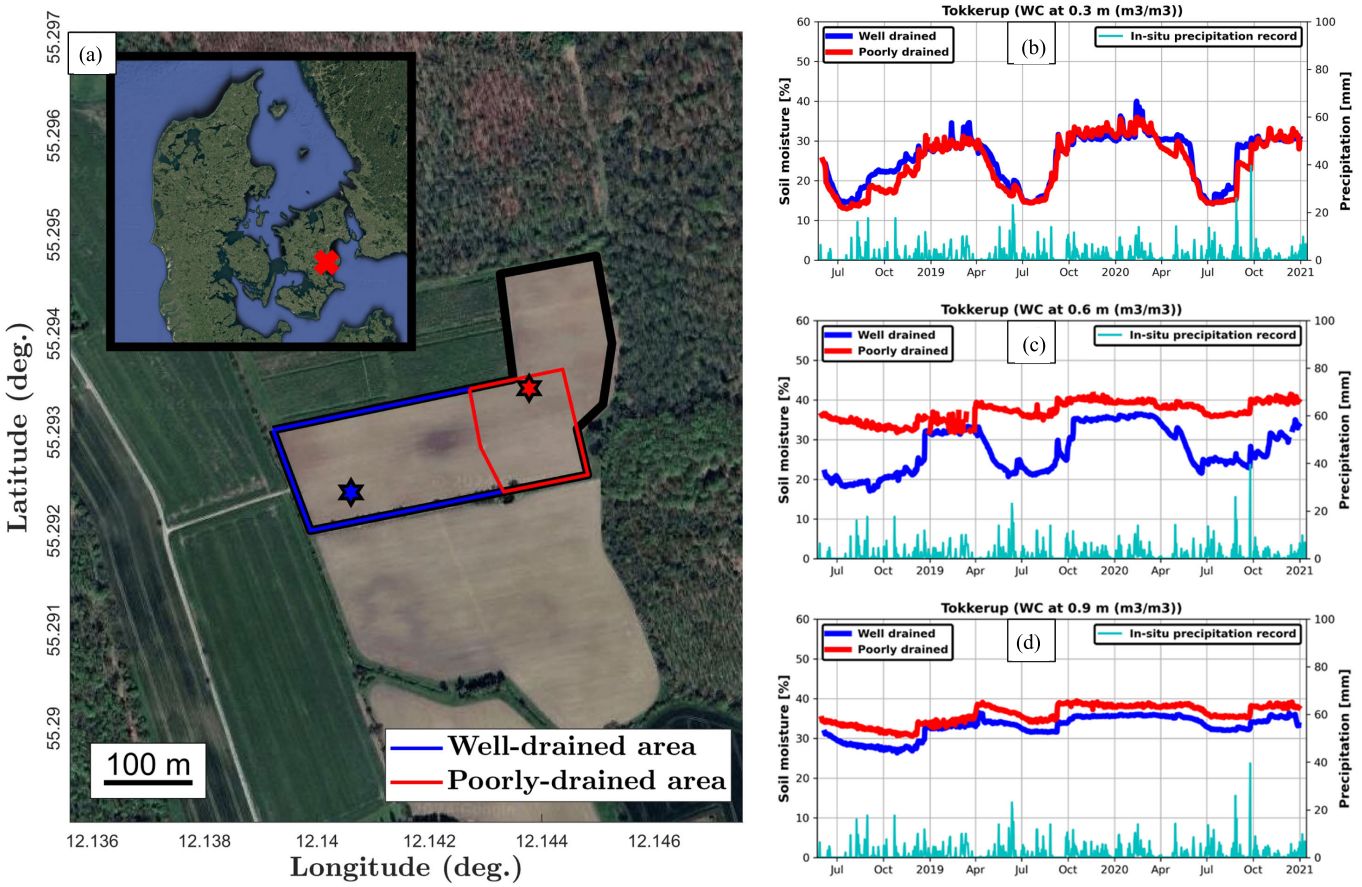


Fig. 4. Study site location, sensor placement, and SM dynamics at Tokkerup, Denmark. (a) Overview map of Denmark indicating the location of the Tokkerup field site (red marker), alongside a satellite image of the field. The field is divided into a WD area (blue outline) and a PD area (red outline). The stars mark the representative locations of the SM sensors within each section. (b)–(d) Volumetric SM content (%) recorded at depths of 0.3, 0.6, and 0.9 m, respectively, from June 2018 to December 2020. SM dynamics for the WD (blue line) and PD (red line) areas are shown alongside in situ precipitation records (cyan bars).

plowing conducted during the ten years leading up to and including the study period (2018–2020). No catch or cover crops were introduced, leaving the soil bare for approximately six months between rotations when WW was followed by SB. Drainage conditions substantially influenced crop growth, particularly during the exceptionally wet autumn–winter season of 2017–2018, which prevented the sowing of WW.

B. HOBE Network

The Danish HOBE was established in 2007 in the Skjern river catchment, western Denmark (see Fig. 5), with the primary objective of achieving water balance closure across multiple spatiotemporal scales through targeted measurements and modeling efforts [24]. As part of this initiative, a dense network of SM monitoring stations was deployed to support hydrological research and satellite-based SM retrievals. These stations have been instrumental in various experimental studies, including the validation of data from the SM and ocean salinity (SMOS) mission under Danish climatic conditions, as detailed in [25].

Within the catchment, 30 SM monitoring stations were strategically arranged in three clusters to optimize spatial coverage and reduce logistical constraints. Stations were numbered according to cluster affiliation, with the first digit indicating the cluster (1)–(3) and the second representing the station's identifier within it (e.g. 3.04). These stations were distributed across various land cover types, including agricultural fields/cropland (22), forests (4), and heathlands (4), providing a comprehensive overview of SM variability across the landscape (see Fig. 5). Nevertheless, since the methodology outlined in this study focuses exclusively on stations located in agricultural fields, only data from these sites will be analyzed.

When the stations were first installed in fall 2009, each was equipped with a Decagon ECH2O data logger connected to three 5TE capacitance sensors, designed to measure SM, temperature, and electrical conductivity at 30-min intervals, later aggregated on an hourly basis. The sensors were positioned horizontally to record data at depth intervals of approximately 0–5, 20–25, and 50–55 cm [25]. In this study, only the 0–5-cm measurements are used, since the penetration depth of the C-band Sentinel-1 radar data is not expected to exceed this range [26].

The HOBE network stations collected data, albeit with some discontinuities, from 2009 to 2019. In agricultural fields, sensors were temporarily removed during cultivation activities, such as seeding, planting, and harvesting. This occurred twice

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

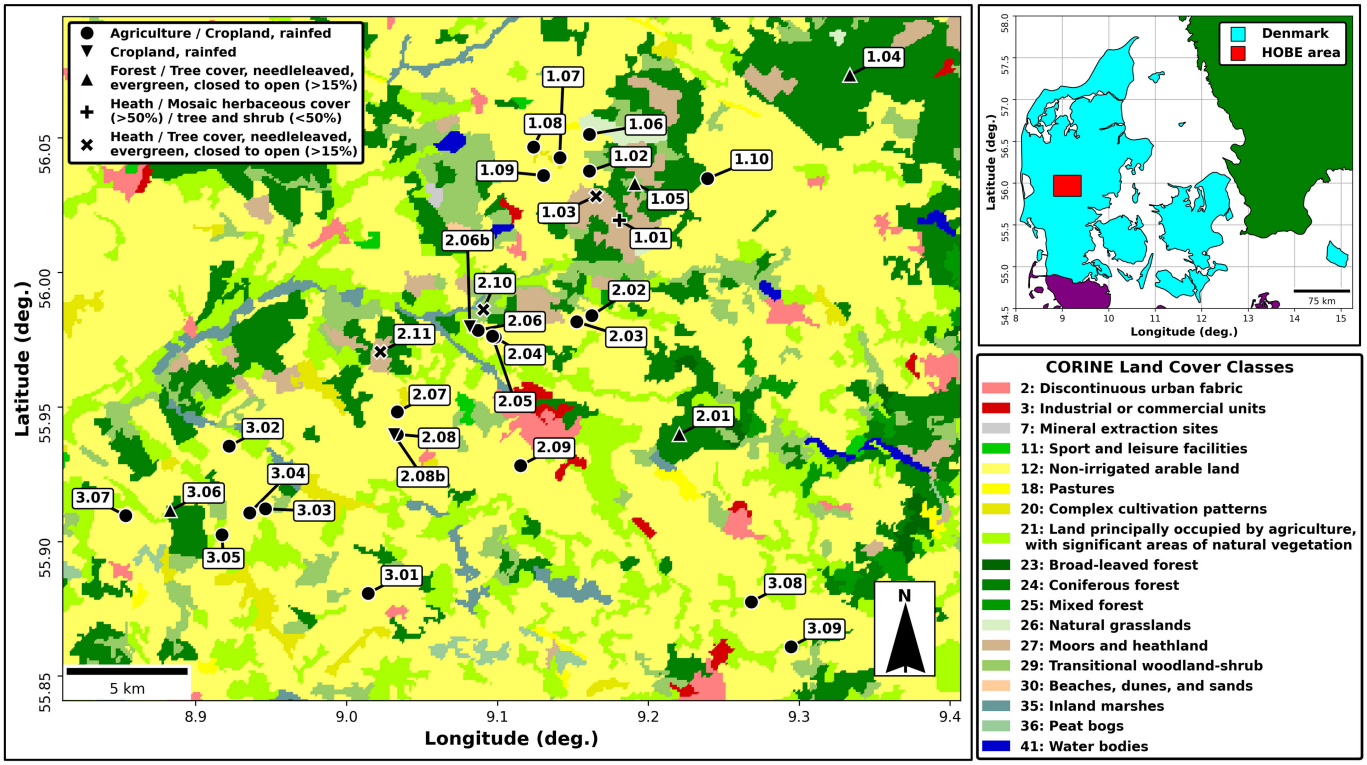


Fig. 5. Distribution of the HOBE SM network stations overlaid on CORINE 2018 land cover data.

a year for summer crops (spring and fall) and once for winter crops (late summer), resulting in occasional measurement gaps [25]. However, this approach was necessary to ensure the data accurately reflected real field conditions. Over time, some sensors required replacement due to maintenance or malfunctions, while certain stations were discontinued altogether. In instances where multiple sensors operated at the same depth, their readings were averaged to improve the representativeness of field moisture conditions.

The 22 agricultural network stations were distributed across fields growing the most common crops in the region: barley, grass, and WW, with additional stations placed in maize and potato fields to account for differences in plant structure [25]. A more detailed analysis of the land cover for each station and year is provided in Table I.

The topsoil in these fields typically consisted of coarse sand, mixed homogeneously with organic material within the plow layer. Beneath this, a pronounced hardpan was usually present at depths of 30–45 cm, limiting the downward movement of water. As a result, sensors placed at 0–5 cm recorded drier conditions due to surface evapotranspiration and rapid infiltration, while those at 20–25 cm showed higher moisture content, positioned just above the hardpan. Bircher et al. [25] emphasize the role of soil texture in moisture retention, noting that soils with higher clay content retained more water, whereas sandy soils exhibited lower moisture levels.

The data collected from these stations are publicly available on the International SM Network (ISMN) website (https://ismn.earth) [27]. Only measurements flagged as "good" by the ISMN's automated quality control system [28] were included in the validation process.

IV. Data Description

This section outlines the datasets necessary to support the retrieval algorithm. These include: Sentinel-1 SAR observations as the primary input; soil texture information to parameterize the electromagnetic model described in Section II-C; coarse resolution SM and field capacity data, used as external constraints in the inversion process outlined in Section II-D. Additionally, geo-referenced field boundaries provided by the Danish Agricultural Agency were employed for backscatter spatial averaging, and crop-type information was used to determine whether the seasonal correction in Section II-E should be applied. Each of these datasets is further detailed in Sections IV-A–IV-D.

A. Sentinel-1 Data

Single look complex (SLC) interferometric wide (IW) Sentinel-1 imagery was processed using GAMMA Remote Sensing software [29]. The processing steps included radiometric calibration, terrain correction, and multilooking (4 times in azimuth and 1 in range), followed by resampling to a 10 × 10-m grid. To reduce noise and capture representative values, the fieldwise median backscatter was computed per scene and used for inversion. For Tokkerup, 680 scenes were analyzed from 2018 to 2020 (both included) using tracks: 044 ascending, 066 descending, 146 ascending, and 168 descending. In the HOBE network, 900 scenes were examined, consisting of tracks: 037 descending, 044 ascending, 066 descending, 117 ascending, and 139 descending. To ensure compatibility with the six-day revisit cycle of the Sentinel-1 constellation, the data validation period was defined

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

TABLE I

STATION WITH RESPECTIVE CROP DATA, GLDAS, AND STCD PERFORMANCE METRICS (R, b, RMSE, AND URMSE), AND SAMPLE SIZE N

Station or network	Crop(s) (2017–2019)	GLDAS				STCD				N
		R	Bias (%)	RMSE (%)	uRMSE (%)	R	Bias (%)	RMSE (%)	uRMSE (%)	
1.06	SW / P / SB	0.82	2.77	4.54	3.60	0.85	1.72	3.72	3.31	136
1.07	PG+C	0.67	2.63	5.75	5.11	0.79	-1.12	4.50	4.36	365
1.08	SB / P / SB	0.80	2.17	3.81	3.13	0.76	-2.60	4.08	3.14	152
1.09	SB / P / SB	0.66	3.33	5.19	3.98	0.66	-1.12	4.02	3.87	183
1.10	SC / SB / P	0.84	8.13	8.56	2.66	0.80	3.42	4.38	2.74	215
2.03	SB / P / SB	0.71	3.16	4.81	3.62	0.67	-1.95	3.60	3.03	88
2.04	SB / SB / P	0.66	0.20	3.64	3.64	0.56	-2.45	5.16	4.54	221
2.05	P / SB / SB	0.71	1.78	3.88	3.45	0.74	-0.74	3.43	3.35	268
2.06b	P / SB / SC	0.68	8.38	9.22	3.83	0.74	2.33	4.11	3.39	231
2.07	SB / SC / SC	0.48	5.46	6.49	3.52	0.48	0.86	3.82	3.73	198
3.01	P / SB / SB	0.59	2.01	5.94	5.59	0.72	-1.91	5.15	4.78	215
3.02	P / SC / SC	0.61	0.78	4.26	4.19	0.68	-0.88	4.59	4.51	254
3.03	SC / SB / SB	-0.04	6.04	7.09	3.70	0.19	3.74	5.02	3.34	57
3.04	WR	0.78	2.18	4.75	4.22	0.78	0.24	4.40	4.40	224
3.05	WB / SB / SB	0.84	1.92	3.20	2.56	0.79	0.66	3.27	3.20	146
3.07	SB / SO / WW	0.82	-0.47	4.71	4.69	0.84	-3.24	5.34	4.24	151
3.08	PG	0.83	3.74	5.41	3.90	0.79	0.53	4.26	4.23	282
3.09	WW / SB / SB	0.76	8.66	9.45	3.77	0.77	-0.30	4.06	4.05	244
HOBE	P-rotation	0.61	3.27	5.82	4.81	0.38	-0.23	6.44	6.43	1963
HOBE	P-rotation	0.61	3.27	5.82	4.81	0.70	-0.33	4.32	4.31	1963
HOBE	No P-rotation	0.64	3.75	6.16	4.88	0.74	-0.17	4.32	4.31	1667
HOBE	All crops	0.63	3.49	5.98	4.85	0.72	-0.26	4.32	4.31	3630
Tokkerup	SB / WW	0.80	-2.01	4.40	3.91	0.69	-4.51	6.79	5.07	467
Tokkerup	SB / WW	0.80	-2.01	4.40	3.91	0.84	-5.62	6.59	3.44	467

Crop abbreviations: P (potatoes), SB (spring barley), SW (spring wheat), SC (silage corn), WB (winter barley), WW (winter wheat), SO (spring oats), PG (permanent grass), C (clover), WR (winter rye). Background color in metric cells reflects performance, with red denoting lower, yellow intermediate, and green higher agreement. Cells highlighted in magenta mark cases where seasonal adjustment was applied.

from 2017 to 2019, inclusive. All processed acquisitions were in VV polarization, chosen for its greater sensitivity to SM variations compared to VH polarization [11].

B. Soil Properties

Ancillary information on sand and clay percentages is required as an input to the selected electromagnetic model [see (9)]. To assess how the choice of soil texture dataset influences the final estimates, three sources with varying coverage and resolution were considered. In situ measurements, often collected alongside SM readings for sensor calibration, provide the most reliable texture data but are spatially limited and may not represent the average conditions within a pixel. They are included here only to evaluate the potential impact of direct measurements on model performance. The DIGIJORD project [30], [31], a Danish initiative integrating satellite data, machine learning, and field observations, offers high-resolution (10 m) national maps of soil properties, including mineral texture fractions for the 0–30-cm layer. While its spatial coverage is limited, its accuracy is likely superior to global products. Finally, OpenLandMap [32] provides global predictions of soil texture fractions and hydrological parameters at a 250-m resolution, with values available at six standard depths. The layer closest to the validation depth was selected. Despite its broad applicability—especially where local datasets are unavailable—its regional accuracy depends on available ground truth. In addition, its simulated surface field capacity at 33 kPa was used as an upper limit to prevent nonphysical SM estimates.

C. Coarse Resolution SM Product

In this study, we use data from the Global Land Data Assimilation System (GLDAS) Noah Land Surface Model L4 3-hourly 0.25° × 0.25° V2.1 (GLDAS_NOAH025_3H), which provides global, gridded estimates of SM at depths of 0–10 cm (surface layer) and 10–40 cm (root zone), updated every 3 h [14]. This dataset is generated using the Noah Land Surface Model, which integrates satellite observations and model-based reanalysis data to simulate land surface processes [33]. The 0.25° spatial resolution (~25 km) supports detailed zonal analysis, providing reasonable initial estimates for our undetermined inversion and acting as a starting point for the proposed retrieval process.

D. LandbrugsGIS: Marker

The Danish Agricultural Agency (Landbrugstyrelsen) provides annual data on agricultural plots in Denmark and the crops grown there. This dataset is created by combining farmers' reports with optical imagery to identify land cover for each field. The shapefiles are open-source

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

and available on the LandbrugsGIS website under "Marker" (https://landbrugsgeodata.fvm.dk/).

V. Results

This section presents the validation results obtained at both test sites using the proposed method. All results are based on an averaging window size of $n = 50$, which was selected as the optimal value for this parameter following the sensitivity analysis detailed in Appendix B.

As described in Section IV, multiple soil texture datasets with varying spatial resolutions were tested within the inversion algorithm to assess their impact on the resulting SM retrievals. However, only marginal differences were observed compared to using in situ measurements, with the DIGIJORD map narrowly outperforming the OpenLandMap model. In the following, DIGIJORD values are used as the reference dataset for all subsequent inversions.

A. Tokkerup

For the study site in Tokkerup, only the near-surface measurements (30 cm depth) were used for validation. Despite being beyond the penetration range of C-band pulses, fluctuations in surface SM are often linked to deeper layers due to vertical water movement [34], [35]. By constraining Sentinel-1 data with coarse-scale estimates modeled at that specific depth and further applying the seasonal adjustment described above, this connection should be better captured and help to bridge the gap between surface observations and deeper soil dynamics by providing physically consistent predictions of moisture redistribution. Since no significant differences were found between the 30-cm measurements in well- and PD patches [see Fig. 4(b)], they were averaged to better represent the field's moisture dynamics.

The model output, overlaid with the recorded in situ measurements, is presented in Fig. 6, with (b) showing the seasonally adjusted retrievals and (a) the uncorrected ones, whereas (c) presents the spatially averaged backscatter. In the latter, annually recurring noise can be observed between July and October, coinciding with strong variability in the normalized difference vegetation index (NDVI) time series derived from Sentinel-2 imagery—possibly due to harvest and sowing activities associated with the SB–WW rotation. Despite the increased signal instability during the summer months, a moderately strong correlation is observed between the backscatter and the measured SM ($R = 0.62$), although a temporal lag is noticeable near the minima (~1 month). This discrepancy may be due to delayed drainage between surface and subsurface soil layers, or more likely, to a seasonal attenuation of the backscatter signal driven by increased vegetation cover during the late growing season.

The final estimates appear to compensate for the observed temporal shift through the application of the seasonal adjustment correction. Overall, the model effectively captures both the dynamics and the amplitude of SM, despite a consistent negative offset of approximately 5%. To understand to what extent the STCD measurements improve compared to the coarse resolution GLDAS product used in the processing,

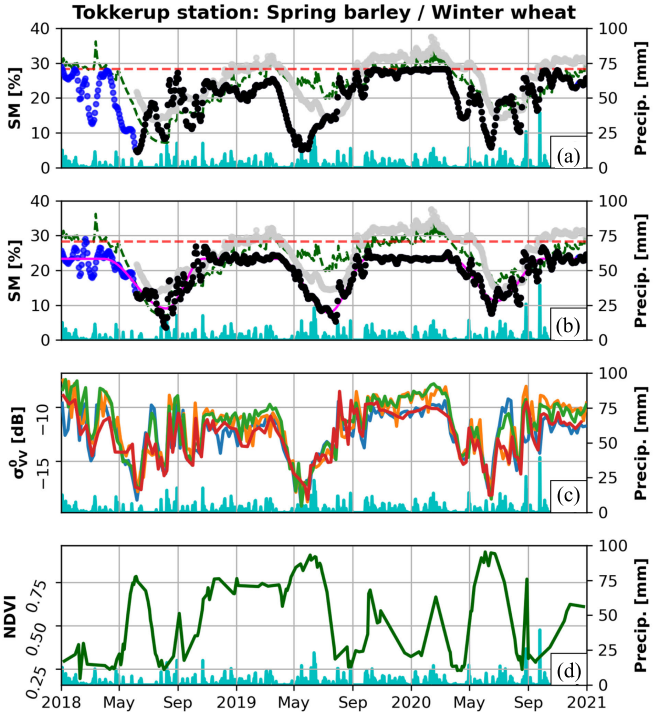


Fig. 6. Time-series analysis of volumetric SM, backscatter, and vegetation dynamics at Tokkerup. (a) and (b) SAR-derived SM retrievals (black and blue dots), with black indicating overlap with in situ measurements (gray line). The uncorrected estimates are shown in (a), while those in (b) underwent seasonal adjustment (in magenta). The red dashed lines mark the field capacity threshold, while the discontinuous green ones correspond to the GLDAS estimates. (c) Spatially averaged backscatter ($\sigma_{0,VV}$) from multiple Sentinel-1 tracks. (d) Sentinel-2 mean NDVI. Cyan bars represent the daily precipitation recorded.

a detailed performance comparison between these estimates is presented in Table I. Multiple statistical metrics were employed, including the Pearson correlation coefficient (R) and the unbiased root mean squared error (uRMSE) (cfr. [36] and (15)–(18) in Appendix B). While the STCD approach yields improvements in R and uRMSE, it does so at the expense of higher bias (b) and overall RMSE. This can be traced down to an underestimation of field capacity in the external dataset, set to 28% and used as an upper bound in the inversion process, highlighting the parameter's influence on retrieval accuracy (see Appendix C). If the field capacity were to be raised to 35%—a plausible value based on the recorded data—the retrieval performance would improve considerably, with metrics reaching $R = 0.88$, RMSE = 4.06%, $b = -1.76\%$, and uRMSE = 3.66%. Nevertheless, the seasonal minima would still tend to fall below the observed values.

Although the current model provides only a single prediction for the entire field, a difference between the well- and PD plots could be observed in the long-term backscatter trends, with the poorly drained areas exhibiting a lower median backscatter level (see Fig. 7). This is unexpected, since higher moisture levels should be associated with higher backscatter. This, however, may be explained by the presence of standing water—typically occurring during episodes of surface saturation—which would in turn cause specular reflection, thus

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

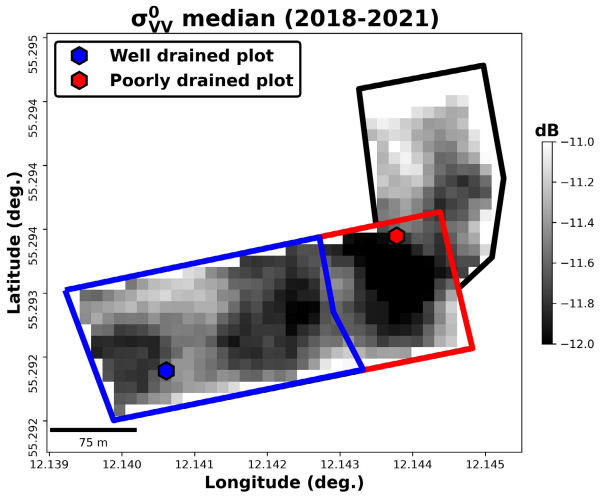


Fig. 7. Median $\sigma_{0,VV}$ backscatter (dB) from Sentinel-1 (2018–2021), with WD (blue) and poorly drained (red) plots outlined.

redirecting the radar pulse away from the sensor and reducing the measured return signal.

B. HOBE Network

Of the 22 stations situated in agricultural areas within the HOBE network, two (2.06 and 2.08) lacked measurements during the timeframe of interest. Additionally, two others were excluded: one (2.08b) because SM measurements consistently exceeded the expected field capacity, suggesting either sensor malfunction or atypical hydrological conditions incompatible with model assumptions; and another (2.09) as it was classified as a natural area, suggesting potentially denser vegetation. After these exclusions, 18 stations remained for further analysis, listed in Table I, alongside their associated evaluation metrics.

The results reflect that the presented methodology generally yields a solid agreement with the measured SM, with most stations exhibiting moderate to high correlation values ($R \geq 0.65$), and RMSE $\leq$ 5%, consistent with similar approaches in [10].

An overview of the aggregated results for both GLDAS and STCD will be provided first, before turning to individual stations and crop-specific patterns. The scatter density plots in Fig. 8 provide a visual comparative evaluation of the GLDAS model and the proposed STCD implementation, against available in situ measurements in all HOBE stations considered. Fig. 8(a) presents the GLDAS model predictions, which show a correlation of $R = 0.63$ and a uRMSE of 4.85%. The model tends to overestimate SM, with predictions clustering between 15% and 25%. Since GLDAS operates at coarse spatial scales, directly comparing its performance to point-based measurements may seem misleading. However, since it serves as a benchmark for our inversion method, this comparison provides valuable insights into how our approach performs relative to an established land surface model.

Fig. 8(b) displays the proposed method retrievals, resulting in higher correlation ($R = 0.72$) and a tighter point redistribution around the truth line (uRMSE = 4.31%), along with reduced RMSE and bias compared to Fig. 8(a). While these enhancements suggest a better fit overall, certain limitations inherited from the coarser dataset persist. Extremes are more accurately captured; however, SM values below 10% are still systematically overpredicted, whereas those exceeding 30% are truncated due to the enforcement of field capacity as an upper bound—a necessary constraint that improves reliability by preventing implausibly high estimates; though it also restricts the model's ability to resolve SM under saturation conditions (such as those following intense rainfall episodes).

Fig. 8(c) and (d) corresponds to the subsets of the stations examined in (a), effectively decomposing it into two groups: those where the seasonal adjustment was not necessary, and those where it was applied due to the presence of tuber or root crops. The resulting metrics—$R = 0.74$ and $R = 0.70$, respectively, with identical uRMSE values of 4.31%—reveal that, upon correction, both groups achieve a comparable level of agreement with in situ measurements.

Finally, Fig. 8(e) and (f) illustrates the relationship between observed backscatter and measured SM for the samples in (c) and (d), both exhibiting a generally positive correlation ($R = 0.69$ and $R = 0.37$). The lower R value in (f) coupled with greater dispersion reinforces the case for recalibrating the final time series based on $S(t)$ to recover the attenuated seasonal signal in the backscatter. Additionally, these findings emphasize that, while backscatter holds valuable information for SM retrieval, further corrections or bounds—such as the use of backscatter ratios and external constraints—are required to reduce the influence of non-SM-related components present in the signal.

Fig. 9 presents the SM time series (in situ, GLDAS, and STCD) for all stations under study. Before proceeding with a more detailed examination, two general considerations are warranted: 1) some pronounced declines and surges are visible during winter—these are most likely associated with freezing events that were not masked in the backscatter data. Yet, their impact on the validation is negligible, as most affected measurements were flagged and excluded from the final evaluation and 2) many of the sharp moisture increases following precipitation episodes are not reflected in the SAR-derived estimates—primarily due to the coarser temporal resolution of the satellite observations, and to a lesser extent, to the imposed upper limit at field capacity. That said, and in order to support a more structured analysis, stations have been grouped into four categories based on land cover characteristics, each of which is discussed in depth in Sections V-B1–V-B4.

1) Potatoes–Cereals: Rotational crop systems involving potatoes—most often paired with SB, though occasionally with silage corn or spring wheat—are the most frequently observed across HOBE stations, present in 10 out of the 18 analyzed. In general, the retrievals show consistent performance after the seasonal adjustment correction is applied, with R values above 0.65 and RMSE below 5%; except for station 2.04, which presents a lower correlation ($R = 0.56$) and a slightly higher RMSE (5.16%). Correlation values across these stations range from 0.56 to 0.84.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

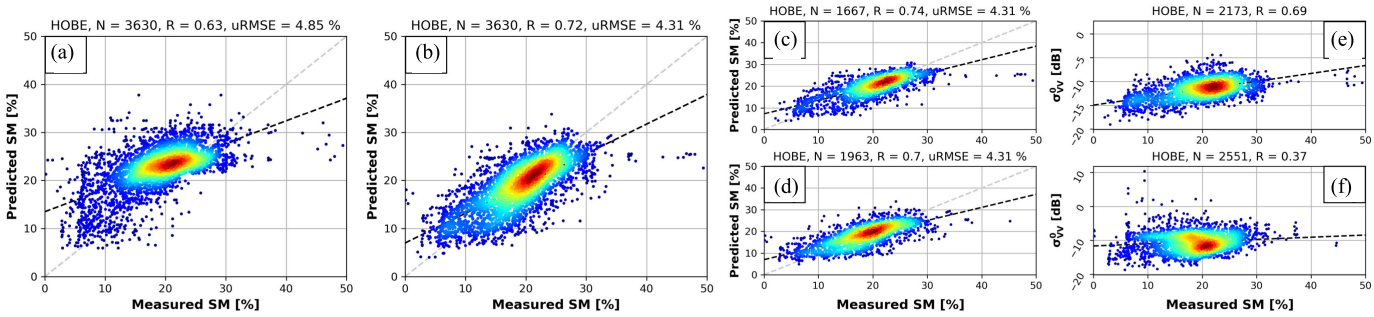


Fig. 8. Scatter density plots comparing predicted versus measured SM at the HOBE network. (a) Performance of the GLDAS model, while (b) presents that of the proposed method. (c) and (d) Results for stations with and without root and tuber crops, respectively; their corresponding backscatter is shown in (e) and (f). Gray dashed lines indicate the 1:1 line, whereas the black dashed ones show linear regression fits.

Several stations—particularly those in the first cluster (e.g., 1.06, 1.08, and 1.09)—exhibit dynamics closely aligned with those predicted by GLDAS, with STCD estimates showing comparable behavior, albeit with some discrepancies or temporal shifts, especially during transitional periods. The most notable example occurs in autumn 2018 at stations 2.04, 2.05, and 3.01, where STCD predicts a smoother change in SM, while GLDAS shows a sudden increase. In the absence of ground-truth data for this timeframe at those stations, it remains unclear whether the gradual change reflects actual conditions or is an artifact introduced by the seasonal adjustment correction, $S(t)$. Nonetheless, the latter explanation appears more plausible, as most SAR-derived time series do exhibit this rapid increase—which would, in turn, be consistent with the expectation that stations within the same catchment feature similar SM dynamics. Moreover, the only two stations with available observations during this interval, 1.07 and 3.08, also display this marked response.

By construction, $S(t)$ introduces smoothness to the seasonal shifts between wet and dry periods, which can at times lead to temporal or magnitude-related misalignments if the parameterization is suboptimal. Nevertheless, even with its shortcomings, the introduction of this correction has proven beneficial for performance at these stations, with the combined R value increasing from 0.38 to 0.70 and RMSE decreasing from 6.44% to 4.31%. These improvements can be difficult to discern when GLDAS and STCD outputs converge but become more noticeable at stations 2.06b and 1.10, where the proposed method captures lower SM levels not reflected in the coarser GLDAS dataset, even though it still tends to overestimate the minima.

2) Silage Corn—SB: Plots featuring both silage corn and SB, without the presence of tuber or root crops during the validation timeframe (namely, stations 2.07 and 3.03), exhibit lower correlation values ($R = 0.48$ and $R = 0.19$, respectively). However, these results should be interpreted with caution: first, because only two stations follow this rotation pattern; and second, due to site-specific factors affecting each case. On the one hand, at station 2.07, the SM estimates follow the observed trends reasonably well, as reflected by the bias, RMSE, and uRMSE values in Table I (all three under 4%). Nonetheless, the measurements span a narrow SM range, making the correlation coefficient highly sensitive to minor deviations from in situ measurements. On the other hand, at station 3.03, the validation data are limited to a single winter period, reducing the sample size to only 57 observations. In this case, however, the statistics do reflect a larger mismatch between model and observations.

3) Permanent Grass and Clover: Permanent grass–clover mixtures (i.e., stations 1.07 and 3.08) provide the longest uninterrupted records, aside from brief disruptions due to freezing events. Station 1.07 spans from January 2017 to March 2019, while station 3.08 covers the period from July 2017 to March 2019. Both produce comparable high-end results relative to the rest of the network, with $R = 0.79$ and RMSE below 5%. Their metrics are of special interest since they provide ground truth across all seasons. While STCD estimates effectively capture lower SM values not reflected in the external prior, they do not reproduce the full extent of the 2018 dry season—and, in the case of station 3.08, neither that of 2017.

4) Winter (–Spring) Cereals: This group comprises stations 3.04, 3.05, 3.07, and 3.09. These are fields that include winter cereals in their crop rotation—typically combined with spring cereals, with station 3.04 being the only exception to that pattern. They all exhibit consistently strong metrics, with $R \geq 0.77$ and RMSE below 4.5%, aside from station 3.07 (5.34%). The model's representation of SM evolution across these sites is particularly robust. Markedly, stations 3.04 and 3.09 reveal distinct declines that are absent in GLDAS data but confirmed by the ground truth. Even so, some discrepancies remain, e.g., in the dry season of 2017. Furthermore, 3.09 reflects a flattened response during winter periods (specially in those of 2018 and 2019) relative to the measured values, possibly due to the imposed field capacity limit.

The consistently strong results observed at stations within the two latter categories may be attributed to structured growth cycles and relatively uniform canopy development, which could arguably reduce vegetation-induced variability in the backscatter signal—thus contributing to a more accurate SM retrieval. Nonetheless, given the limited number of stations representing these crop types and the difficulty of evaluating their diverse rotational practices, drawing definitive conclusions about their specific influence on SAR-based SM estimation remains challenging. This consideration can be extended to all other groups addressed in this section.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

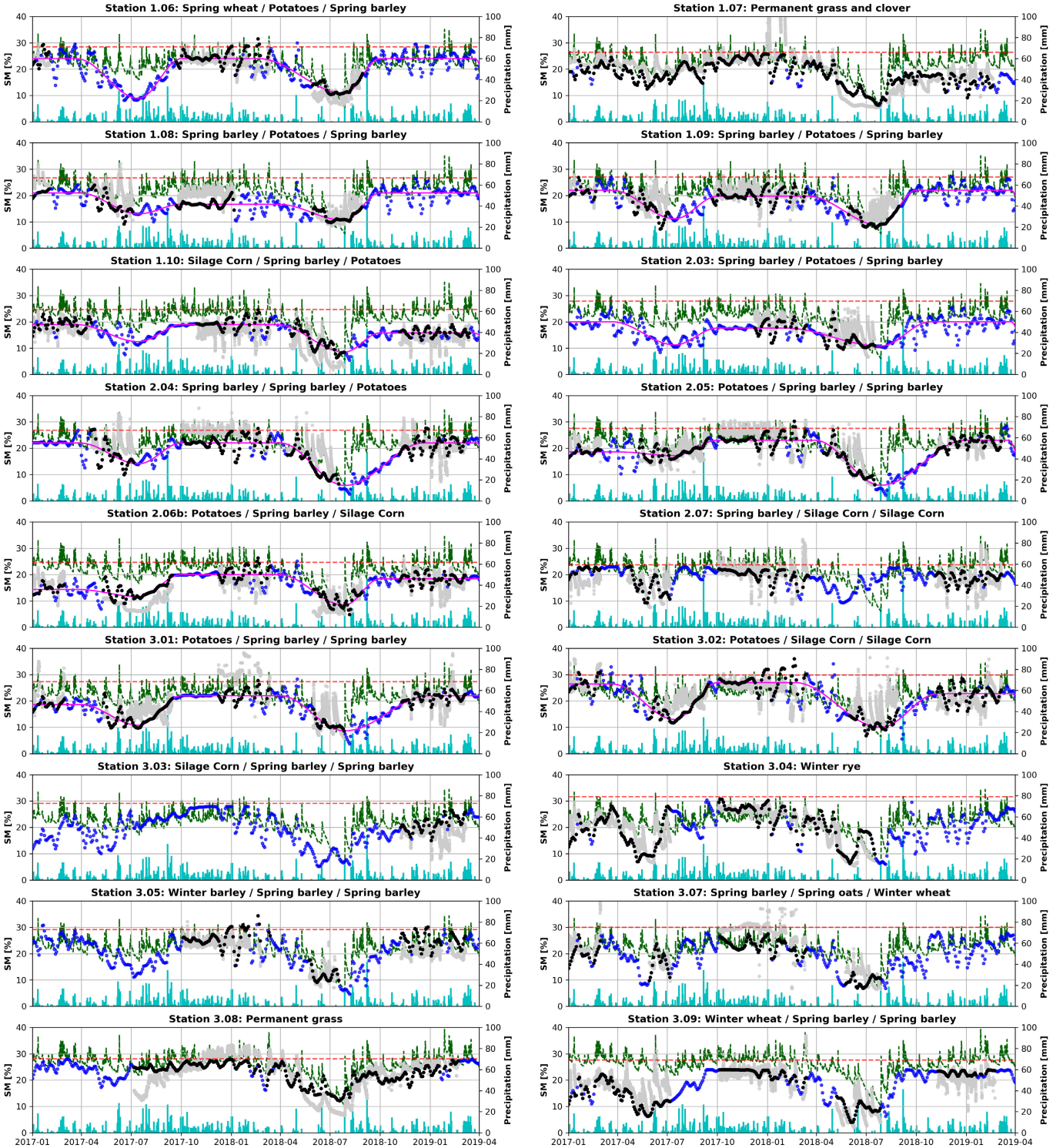


Fig. 9. SM time series at the 18 HOBE stations analyzed (2017–2019). Black and blue dots show SAR-derived SM retrievals, with black indicating overlap with in situ observations (gray). The magenta lines represent the seasonal adjustment curve $S(t)$, while the red dashed ones mark the field capacity threshold. The discontinuous green lines correspond to the GLDAS estimates. Panel titles state station identifier and corresponding crop types. Daily precipitation records (cyan bars) were obtained from the DMI Open Data `climateData` API (10 km grid values) [37].

VI. Conclusion

A constrained STCD methodology was successfully implemented and validated against multiple test sites in Denmark. The impact of key algorithm parameters, such as the spatial averaging and the temporal window size used in the inversion, was investigated. A resampling grid of 10×10 m was chosen following the application of multilooking factors of 4 times in azimuth and 1 in range. For each field of interest, the median backscatter value of all grid pixels within its boundaries was computed and used as the primary input for the inversion

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

process. Further investigation—supported by multiple, spatially distinct in-field measurements—would be required to assess the feasibility of estimating SM at the subfield scale.

An averaging window size of $n = 50$ was selected based on empirical simulations evaluating the model's performance. As previously discussed, the lower values of n further constrain the backscatter time series to the coarser external data estimates, thereby reducing variability. In contrast, the larger values of n improve spatial detail by allowing the signal to reflect finer-scale features but also increase the risk of capturing backscatter effects not driven by SM. At stations where this issue arose, a seasonal correction was applied using a simplified model mimicking the expected SM dynamics over the annual cycle. Nonetheless, in some instances, transitions between wet and dry periods, or vice versa, were overly smoothed by these adjustments, likely due to suboptimal parametrization. Possible improvements could involve coupling the parameters with hydrological models or rainfall records. This correction could alternatively be circumvented if additional preprocessing steps are applied to the Sentinel-1 data to mitigate signal contributions unrelated to SM fluctuations, as described in [10] (e.g., vegetation correction). Along these lines, several strategies have been proposed in literature, including leveraging simple backscatter/polarization decompositions to reduce volume-scattering contributions and incorporating optical proxies (e.g., NDVI, LAI) to down-weight periods of rapid canopy change through soil and vegetation scattering modeling [8], [38], [39].

Overall, the measured SM time series was found to reproduce the highly nonlinear patterns observed in the ground-truth data—in some cases not present in the coarser dataset—albeit for some outliers, which were mainly located either in stations where the recorded SM systematically surpassed the expected field capacity, or in stations where the STCD method main assumptions do not consistently hold true. No significant differences in accuracy were observed between stations where the seasonal correction was applied—either due to the depth of the validation measurements or the presence of tuber/root crops—and those where it was not. That said, the lack of data during transitional periods at some HOBE stations may have obscured potential underperformance at sites where the correction was applied.

Multiple external data sources were used to derive the final estimates, including three different soil texture datasets with varying resolutions and coverages. Their assessment revealed that these variations had minimal impact on the final model performance, suggesting that their influence is less critical than initially expected. This does not imply that soil texture itself is unimportant for moisture retention but rather that the algorithm's predictions remain largely unaffected by minor variations in this type of data, highlighting a potential area for improvement.

In contrast, field capacity as an upper boundary plays a more influential role, as illustrated by the SM retrieval in Tokkerup. Predicting high SM levels (above 30%) remains challenging, as most OpenLandMap field capacity estimates fall below this threshold. This can be appreciated from the flat distribution of data points in Fig. 8(b) within the SM measured range of 30%–40%. Yet, loosening this constraint results in an excessive number of outliers, revealing the tradeoff between capturing higher SM values and maintaining good prediction reliability (see Appendix C).

The presented approach is mainly limited by two factors: its main assumption that other contributors to the radar return occur at longer timescales than surface SM changes—which does not always hold—and its reliance on coarser SM models. Both limitations could potentially be addressed through the integration of complementary hydrological datasets or ancillary information characterizing land surface processes, such as vegetation dynamics, surface roughness changes, or freeze–thaw cycles.

Despite these caveats, the method offers several practical advantages. Its strengths lie in its simplicity and global applicability, providing fieldwise estimates when applied in regions where the model's core assumptions are valid. Its flexibility in switching between external datasets based on regional availability further enhances its adaptability. Most notably, the results have shown that even when the coarser external SM range does not fully capture actual variations, the backscatter measurements may, in some cases, help compensate for these discrepancies—provided that the temporal dynamics are well represented.

Appendix A
Prior Selection (α_0)

One of the main distinctions of the methodology presented here, relative to the original formulation of the STCD approach, [19], is how the undetermined system (5) is solved.

In [3], the system is completed via a maximum-likelihood estimator, computed from the minimum polarization amplitude within that window ($\alpha_{\min}$), which in turn estimated using a coarser SM dataset. For the jth window of size n, the prior is then

$$\alpha_0 = \alpha_{\min} \cdot \max\left(\sqrt{\frac{\sigma_{0,\mathrm{VV}}^{(j+n-1)}}{\vec{\sigma}_{0,\mathrm{VV}}^{\mathrm{win}}}}\right) \tag{13}$$

and from there, it is possible to show that

$$|\vec{\alpha}_{VV}| = \alpha_0 \cdot \sqrt{\frac{\vec{\sigma}_{0,\mathrm{VV}}^{\mathrm{win}}}{\sigma_{0,\mathrm{VV}}^{(j+n-1)}}} = \alpha_{\min} \cdot \sqrt{\frac{\vec{\sigma}_{0,\mathrm{VV}}^{\mathrm{win}}}{\min\left(\vec{\sigma}_{0,\mathrm{VV}}^{\mathrm{win}}\right)}}. \tag{14}$$

This approach thus effectively enforces a lower bound for $|\vec{\alpha}_{\mathrm{VV}}|$ at $\alpha_{\min}$ by construction.

In practice, this procedure was found to be highly sensitive to the fidelity of the external product in capturing the true window-specific minimum. For the cases considered in this article, the GLDAS product [14] systematically overestimated lower SSM values, which would therefore introduce a bias in the retrievals, regardless of the window size employed, as shown in Fig. 10 for the HOBE test site.

To address this issue, an alternative constraint is proposed, according to which the relation between the normalized backscattering coefficient and volumetric SM is assumed to be linear within each estimation window. The prior α_0 is then

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

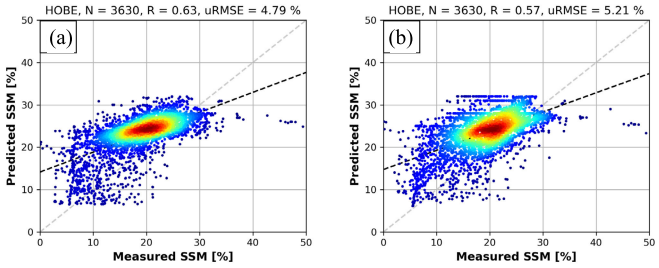


Fig. 10. Scatter-density plots of predicted versus measured SM across the HOBE network. (a) and (b) Compute α_0 following [3], with $n = 4$ (as recommended therein) and $n = 50$ (optimal time-window size identified in this study; see Appendix B), respectively.

obtained from the SM value corresponding to the last date in the estimation window, Θ_0, which is computed through (11). The estimate is then propagated backward to the remaining acquisitions in the window, allowing the final retrieval to extend beyond the range spanned by the external product for the same timeframe. When the constraint provided by (11) is used together with a sufficiently large estimation window (see Appendix B) and under all the assumptions discussed for the STCD approach, it is sufficient for the external dataset to capture the average level of the SM within the estimation window to allow for accurate retrievals.

Appendix B
Optimal Time-Window Size n

To determine the optimal time-window size for the inversion algorithm described in Section II (see Fig. 2), SM retrieval simulations were conducted and tested against observations from the 18 HOBE stations selected for validation between 2017 and 2019. Each integer value of n from 2 to 100 was evaluated. For reference, $n = 1$ corresponds to yielding estimates entirely relying in the coarser GLDAS dataset, sampled at the Sentinel-1 acquisition times, without incorporating any backscatter information. Each unit increase in n extends the window timeframe by six days (revisit time for a given track).

To quantify how changes in n affect retrieval accuracy, the following metrics were used

$$b = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{\Theta}_i - \Theta_i\right) \tag{15}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{\Theta}_i - \Theta_i\right)^2} \tag{16}$$

$$\text{uRMSE} = \sqrt{\text{RMSE}^2 - b^2} \tag{17}$$

$$R = \frac{\sum_{i=1}^{N}\left(\Theta_i - \bar{\Theta}\right)\left(\hat{\Theta}_i - \bar{\hat{\Theta}}\right)}{\sqrt{\sum_{i=1}^{N}\left(\Theta_i - \bar{\Theta}\right)^2}\sqrt{\sum_{i=1}^{N}\left(\hat{\Theta}_i - \bar{\hat{\Theta}}\right)^2}}. \tag{18}$$

The results of this parameter sweep are portrayed in Fig. 11. Fig. 11(a)–(d) shows a clear minimum around $n = 50$ for uRMSE, RMSE, and the absolute bias, $|b|$, as R simultaneously reaches its peak value. Despite some variability across stations,

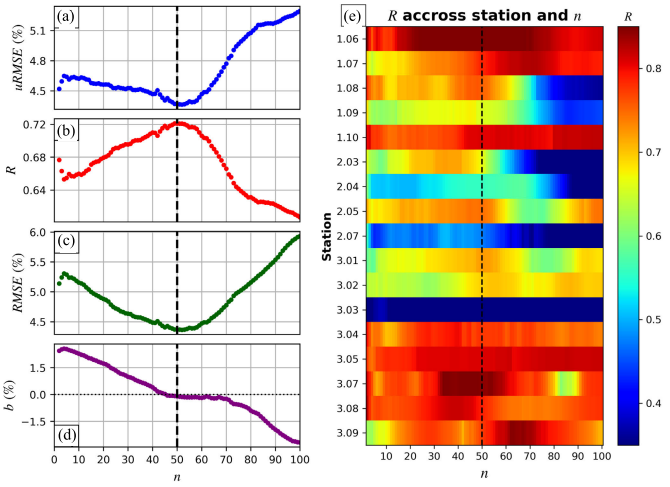


Fig. 11. Sensitivity analysis of the average window size (n) on retrieval performance. (a)–(d) Combined performance for all stations considered as a function of n: (a) unbiased RMSE (uRMSE), (b) Pearson correlation coefficient (R), (c) RMSE, and (d) bias (b). (e) R values across all stations and window sizes. The vertical dashed line marks the window size selected as optimal ($n = 50$).

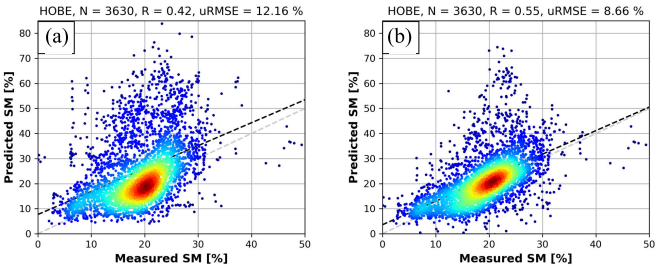


Fig. 12. (a) and (b) Performance of the proposed method without enforcing a field capacity upper limit, where the seasonal adjustment correction is omitted in the former and applied in the latter. Only stations where tuber or root crops are present in the rotational system undergo this last processing step.

Fig. 11(e) also indicates that the highest correlation values tend to cluster around intermediate values of n within the explored range. Based on these results, $n = 50$ was selected as the default value for the proposed inversion method, corresponding to a 300-day window that roughly aligns with the annual cycle, thereby potentially leveraging seasonal variations in backscatter, if present.

Appendix C
Impact of Field Capacity

To assess the influence of enforcing field capacity as an upper limit in the STCD retrieval, a comparison was performed using SM observations across the selected HOBE stations, with and without the application of this constraint. Fig. 12 presents the results when no field capacity threshold is applied. In the absence of this ceiling value, the method tends to overestimate SM, especially under wet conditions, producing unphysically high predictions, particularly in stations where tuber or root crops are present and vegetation-related effects may distort the backscatter signal. The application of the seasonal adjustment where appropriate, as depicted in Fig. 12(b),

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

partially reduces the number of outliers, improving correlation (R = 0.55 versus 0.42) and lowering uRMSE (8.66% versus 12.16%). Nonetheless, this correction effectively introduces an implicit upper bound, as $S(t)$ sets the winter maxima to values no higher than 5% below field capacity (refer to Section II-E).

In contrast, Fig. 8(b) shows the performance when a field capacity limit is enforced. This constraint eliminates extreme overestimates and improves retrieval consistency across the full dynamic range. The resulting STCD estimates achieve a substantially higher overall correlation (R = 0.72) and lower uRMSE (4.31%), indicating a marked improvement in retrieval quality. These results underscore the importance of applying physically meaningful bounds when backscatter ratios alone are insufficient to reliably characterize moisture conditions.

ACKNOWLEDGMENT

The SM data from the Tokkerup site, along with the DIGIJORD soil texture maps, were provided by the Department of Agroecology at Aarhus University as part of the KortDrænN2O project.

REFERENCES

[1] R. A. Lane, V. A. Bell, R. M. Chapman, and A. L. Kay, "Evaluating soil moisture simulations from a national-scale gridded hydrological model over Great Britain," *J. Hydrol., Regional Stud.*, vol. 52, Apr. 2024, Art. no. 101735. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2214581824000831

[2] T. N. Carlson and G. P. Petropoulos, "A new method for estimating of evapotranspiration and surface soil moisture from optical and thermal infrared measurements: The simplified triangle," *Int. J. Remote Sens.*, vol. 40, no. 20, pp. 7716–7729, Oct. 2019.

[3] A. Balenzano et al., "Sentinel-1 soil moisture at 1km resolution: A validation study," *Remote Sens. Environ.*, vol. 263, Sep. 2021, Art. no. 112554. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425721002741

[4] Y. H. Kerr et al., "The SMOS mission: New tool for monitoring key elements ofthe global water cycle," *Proc. IEEE*, vol. 98, no. 5, pp. 666–687, May 2010.

[5] W. Wagner et al., "ASCAT soil moisture calibration using in situ data from the international soil moisture network," *Int. J. Remote Sens.*, vol. 34, nos. 9–10, pp. 3521–3543, 2013.

[6] S. Chan et al., "Development and assessment of the SMAP enhanced passive soil moisture product," *Remote Sens. Environ.*, vol. 204, pp. 931–941, Jan. 2017.

[7] J. Peng et al., "A roadmap for high-resolution satellite soil moisture applications—Confronting product characteristics with user requirements," *Remote Sens. Environ.*, vol. 252, pp. 112–162, Jan. 2021. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425720305356

[8] D. Fan et al., "A Sentinel-1 SAR-based global 1-km resolution soil moisture data product: Algorithm and preliminary assessment," *Remote Sens. Environ.*, vol. 318, Mar. 2025, Art. no. 114579. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425724006059

[9] R. Meyer et al., "Exploring the combined use of SMAP and Sentinel-1 data for downscaling soil moisture beyond the 1 km scale," *Hydrol. Earth Syst. Sci.*, vol. 26, no. 13, pp. 3337–3357, Jul. 2022. [Online]. Available: https://hess.copernicus.org/articles/26/3337/2022/

[10] D. Mengen, T. Jagdhuber, A. Balenzano, F. Mattia, H. Vereecken, and C. Montzka, "High spatial and temporal soil moisture retrieval in agricultural areas using multi-orbit and vegetation adapted Sentinel-1 SAR time series," *Remote Sens.*, vol. 15, no. 9, p. 2282, Apr. 2023. [Online]. Available: https://www.mdpi.com/2072-4292/15/9/2282

[11] T. Ma, L. Han, and Q. Liu, "Retrieving the soil moisture in bare farmland areas using a modified Dubois model," *Frontiers Earth Sci.*, vol. 9, pp. 1–14, Dec. 2021. [Online]. Available: https://www.frontiersin.org/journals/earth-science/articles/10.3389/feart.2021.735958

[12] A. G. Voronovich, *Wave Scattering From Rough Surfaces* (Springer Series on Wave Phenomena), vol. 17. Berlin, Germany: Springer-Verlag, 1994, doi: 10.1007/978-3-642-97544-8.

[13] M. Hallikainen, F. Ulaby, M. Dobson, M. El-Rayes, and L.-K. Wu, "Microwave dielectric behavior of wet soil—Part 1: Empirical models and experimental observations," *IEEE Trans. Geosci. Remote Sens.*, vol. GE-23, no. 1, pp. 25–34, Jan. 1985.

[14] H. Beaudoing and M. Rodell, "GLDAS Noah land surface model L4 3 hourly 0.25 × 0.25 degree; version 2.1," NASA Goddard Earth Sci. Data Inf. Services Center (GES DISC), Greenbelt, MD, USA, 2020. Accessed: Feb. 2, 2025. [Online]. Available: https://doi.org/10.5067/E7TYRXPJKWOQ

[15] Z. Bartalis, V. Naeimi, S. Hasenauer, and W. Wagner, "ASCAT soil moisture product handbook," Inst. Photogramm. Remote Sens., Vienna Univ. Technol., Vienna, Austria, Tech. Rep. 15, 2008.

[16] S. K. Chan et al., "Assessment of the SMAP passive soil moisture product," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4994–5007, Aug. 2016.

[17] Y. H. Kerr, P. Waldteufel, J.-P. Wigneron, J. Martinuzzi, J. Font, and M. Berger, "Soil moisture retrieval from space: The Soil Moisture and Ocean Salinity (SMOS) mission," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 8, pp. 1729–1735, Aug. 2001.

[18] E. G. Njoku, T. J. Jackson, V. Lakshmi, T. K. Chan, and S. V. Nghiem, "Soil moisture retrieval from AMSR-E," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 2, pp. 215–229, Feb. 2003.

[19] A. Balenzano et al., "On the use of temporal series of L-and X-band SAR data for soil moisture retrieval. Capitanata plain case study," *Eur. J. Remote Sens.*, vol. 46, no. 1, pp. 721–737, Jan. 2013, doi: 10.5721/eujrs20134643.

[20] M. Arias, C. Notarnicola, M. Á. Campo-Bescós, L. M. Arregui, and J. Álvarez-Mozos, "Evaluation of soil moisture estimation techniques based on Sentinel-1 observations over wheat fields," *Agricult. Water Manage.*, vol. 287, Sep. 2023, Art. no. 108422. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0378377423002871

[21] S. Khabbazan et al., "Crop monitoring using Sentinel-1 data: A case study from The Netherlands," *Remote Sens.*, vol. 11, no. 16, p. 1887, Aug. 2019. [Online]. Available: https://www.mdpi.com/2072-4292/11/16/1887

[22] S. K. Motarjemi, M. E. Styczen, R. J. Petersen, K. J. S. Jensen, and F. Plauborg, "Effects of different drainage conditions on nitrogen losses of an agricultural sandy loam soil," *J. Environ. Manage.*, vol. 325, Jan. 2023, Art. no. 116267. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0301479722018400

[23] O. H. Jacobsen and P. Schjønning, "A laboratory calibration of time domain reflectometry for soil water measurement including effects of bulk density and texture," *J. Hydrol.*, vol. 151, nos. 2–4, pp. 147–157, Nov. 1993. [Online]. Available: https://www.sciencedirect.com/science/article/pii/002216949390233Y

[24] K. H. Jensen and J. C. Refsgaard, "HOBE: The Danish hydrological observatory," *Vadose Zone J.*, vol. 17, no. 1, pp. 1–24, Jan. 2018, doi: 10.2136/vzj2018.03.0059.

[25] S. Bircher, N. Skou, K. H. Jensen, J. P. Walker, and L. Rasmussen, "A soil moisture and temperature network for SMOS validation in Western Denmark," *Hydrol. Earth Syst. Sci.*, vol. 16, no. 5, pp. 1445–1463, May 2012. [Online]. Available: https://hess.copernicus.org/articles/16/1445/2012/

[26] H. Bazzi et al., "Retrieving soil moisture from Sentinel-1: Limitations over certain crops and sensitivity to the first soil thin layer," *Water*, vol. 16, no. 1, p. 40, 2023. [Online]. Available: https://www.mdpi.com/2073-4441/16/1/40

[27] W. A. Dorigo et al., "The international soil moisture network: A data hosting facility for global in situ soil moisture measurements," *Hydrol. Earth Syst. Sci.*, vol. 15, no. 5, pp. 1675–1698, May 2011. [Online]. Available: https://hess.copernicus.org/articles/15/1675/2011/

[28] W. A. Dorigo et al., "Global automated quality control of in situ soil moisture data from the international soil moisture network," *Vadose Zone J.*, vol. 12, no. 3, pp. 1–21, Aug. 2013, doi: 10.2136/vzj2012.0097.

[29] C. Werner, U. Wegmüller, T. Strozzi, and A. Wiesmann, "Gamma SAR and interferometric processing software," in *Proc. ERS-ENVISAT Symp.*, Oct. 2000, pp. 1–9. [Online]. Available: https://api.semanticscholar.org/CorpusID:28598270

[30] A. B. Møller, M. H. Greve, and A. M. Beucher, "Opdateret jordbundstypekort," DCA–Nationalt Center Fødevare Og Jordbrug, Aarhus Universitet, Tech. Rep. 2023-0626709, Mar. 2024.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.

[31] A. B. Møller et al., "High-resolution 3D soil texture mapping in Denmark using satellite time series and bare soil composites," *Adv. Agronomy*, vol. 194, pp. 133–186, Jan. 2025.
[32] T. Hengl et al., "SoilGrids250m: Global gridded soil information based on machine learning," *PLoS ONE*, vol. 12, no. 2, pp. 1–40, Feb. 2017, doi: 10.1371/journal.pone.0169748.
[33] G.-Y. Niu et al., "The community Noah land surface model with multiparameterization options (Noah-MP): 1. Model description and evaluation with local-scale measurements," *J. Geophys. Res.*, vol. 116, no. D12, pp. 1–19, 2011, doi: 10.1029/2010jd015139.
[34] C. Albergel et al., "From near-surface to root-zone soil moisture using an exponential filter: An assessment of the method based on in-situ observations and model simulations," *Hydrol. Earth Syst. Sci.*, vol. 12, no. 6, pp. 1323–1337, Dec. 2008. [Online]. Available: https://hess.copernicus.org/articles/12/1323/2008/
[35] L. Brocca et al., "Soil moisture estimation through ASCAT and AMSR-E sensors: An intercomparison and validation study across Europe," *Remote Sens. Environ.*, vol. 115, no. 12, pp. 3390–3408, Dec. 2011. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425711002756
[36] A. Gruber et al., "Validation practices for satellite soil moisture retrievals: What are (the) errors?," *Remote Sens. Environ.*, vol. 244, Jul. 2020, Art. no. 111806. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425720301760
[37] Danish Meteorological Institute (DMI). *DMI Open Data: Climate Data API (ClimateData)-10kmGridValue*. Data Distributed Under CC BY 4.0. Accessed: Dec. 5, 2025. [Online]. Available: https://www.dmi.dk/friedata/dokumentation/climate-data
[38] L. Mascolo, S. R. Cloude, and J. M. Lopez-Sanchez, "Model-based decomposition of dual-pol SAR data: Application to Sentinel-1," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5220119.
[39] S. Nativel et al., "Hybrid methodology using Sentinel-1/Sentinel-2 for soil moisture estimation," *Remote Sens.*, vol. 14, no. 10, p. 2434, May 2022. [Online]. Available: https://www.mdpi.com/2072-4292/14/10/2434



Miquel Negre Dou received the B.Sc.Eng. degree in engineering physics from the Polytechnic University of Catalonia (UPC), Barcelona, Spain, in 2020, and the M.Sc.Eng. degree in Earth and space physics and engineering from the Technical University of Denmark (DTU), Kongens Lyngby, Denmark, in 2023, where he is currently pursuing the Ph.D. degree with the Microwaves and Remote Sensing Division, National Space Institute, with a focus on soil moisture estimation using synthetic aperture radar (SAR) observables.



John Peter Merryman Boncori received the Laurea degree (summa cum laude) in communications engineering and the Ph.D. degree in geoinformation from the University of Rome "Tor Vergata," Rome, Italy, in 2003 and 2007, respectively.

Since then, he has been working as a Researcher with the Technical University of Denmark, Kongens Lyngby, Denmark, and the National Institute of Geophysics and Volcanology, Rome. Since 2018, he has been an Associate Professor with the Microwaves and Remote Sensing Division, DTU Space, Technical University of Denmark, where he has been the Head of Division, since 2020. He has worked on a range of international research projects related to SAR data processing and interferometric synthetic aperture radar (SAR) techniques and their application in measurements of, e.g., seismic deformation and ice flow.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:18:40 UTC from IEEE Xplore. Restrictions apply.